Tarkan Zarrouk

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SUMMARY

Mathematics student at Carleton University with a strong interest in pure mathematics, particularly mathematical analysis, abstract algebra, and rigorous proof-based reasoning. Experienced in frontend development, database integration, and software engineering through startup work and personal projects. Proficient in React, TypeScript, JavaScript, and Python, with an interest in exploring the connections between mathematical structures, computational logic, and programming.

EXPERIENCE

Frontend Web Developer

Kaie **February 2025 - April 2025, Mississauga, Ontario, Canada**

- **Led the implementation of real-time collaborative editing**, enabling multiple users to edit shared content simultaneously and improving coordination across collaborative workflows.
- **Designed, developed, and shipped 2 major user-facing features**, user profile management and an interactive Explore page, delivering responsive interfaces aligned with evolving product requirements.
- **Built and refined reusable frontend components** and optimized key UI workflows, improving interface consistency, simplifying future feature development, and enhancing maintainability.
- **Migrated 2,000+ user records to a new database**, supporting data integrity through careful transfer and validation while minimizing disruption during the transition.
- **Contributed across the full development lifecycle**, collaborating with teammates to implement, test, debug, and deploy features across releases in a fast-paced startup environment.

Sensei

Code Ninjas **July 2024 - August 2024, Cooksville, Mississauga, Ontario, Canada**

- Instructed students aged 6–13 in programming fundamentals, adapting lessons to different age groups and skill levels.
- Taught computational thinking, logical reasoning, and debugging through hands-on coding activities and practical exercises.
- Managed classroom environments of up to 30 students and instructors, maintaining engagement and supporting collaborative learning.
- Collaborated with 2–4 colleagues to coordinate lessons, support students, and deliver effective programming instruction.
- Developed and implemented a curriculum enhancement for advanced students, incorporating game development principles and project-based learning modules.

PROJECT

Linear Algebra Visualization App

- Interactive linear algebra visualizer for Carleton University MATH 1104 students.
- Visualizes matrices, vectors, linear systems, and transformations in 2D and 3D.
- Step-by-step solutions to connect calculations with exam techniques.
- Designed to strengthen conceptual understanding through interactive exploration.
- Enhanced accuracy across solution techniques, modern concepts, and visualization.

Social Media App

- Developed a full-stack social media application using **Next.js, React, and TypeScript**, building **5+ core areas** including the social feed, user profiles, post management, user discovery, and account settings.
- Implemented **3 key post capabilities**: text and image attachments, likes, and bookmarks, with character-limit validation and persistent data storage through Firebase Firestore.
- Built user discovery and networking features supporting **3 core interactions**: searching for users, viewing profiles, and following or unfollowing accounts, with changes persisted to Firestore.
- Integrated Firebase Authentication across **5+ account workflows**, including registration, sign-in, password recovery, email verification, and account management; also implemented profile editing and email and password updates.

- Developed reusable React components and responsive interfaces using **Tailwind CSS**, structuring the application across **8+ dedicated page routes** for feeds, search, bookmarks, profiles, authentication, and settings.

EDUCATION

Honours Bachelor in Mathematics

Carleton University • Ottawa, Ontario, Canada

- Relevant Coursework: Linear Algebra, Multivariable Calculus, Mathematical Analysis, Abstract Algebra
- Developing a foundation in mathematical reasoning, formal proofs, and analytical problem-solving.
- Exploring the intersection of mathematics and programming, with interests in theoretical mathematics and computational applications.

High School Diploma

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- Graduated with a 3.56/4.00 GPA while participating in technology-focused extracurricular activities.
- Participated in the Robotics Team and Computer Science Club, developing technical and collaborative problem-solving skills.
- Competed in Skills Canada Web Design, Ontario Tech Engineering Robotics, Skills Ontario Robotics, and the Waterloo CSIMC mathematics contest.
- Participated in the Information and Communications Technology (ICT) program, gaining exposure to programming, technology, and applied computing.

SKILLS

Languages: JavaScript, TypeScript, Python, Java, C++, HTML5, CSS3
Frontend Development: React, Next.js, Tailwind CSS, Responsive UI Design, Component-Based Architecture
Backend & Databases: Firebase, CRUD Operations, Database Integration, Data Validation
Computer Vision & AI: OpenCV, YOLOv5, MediaPipe, Image Processing, Object Detection
Embedded Systems: Arduino, Ultrasonic Sensors, Motor Control, Hardware-Software Integration
Tools & Practices: Git, GitHub, Debugging, Testing, Troubleshooting, Modular Software Design